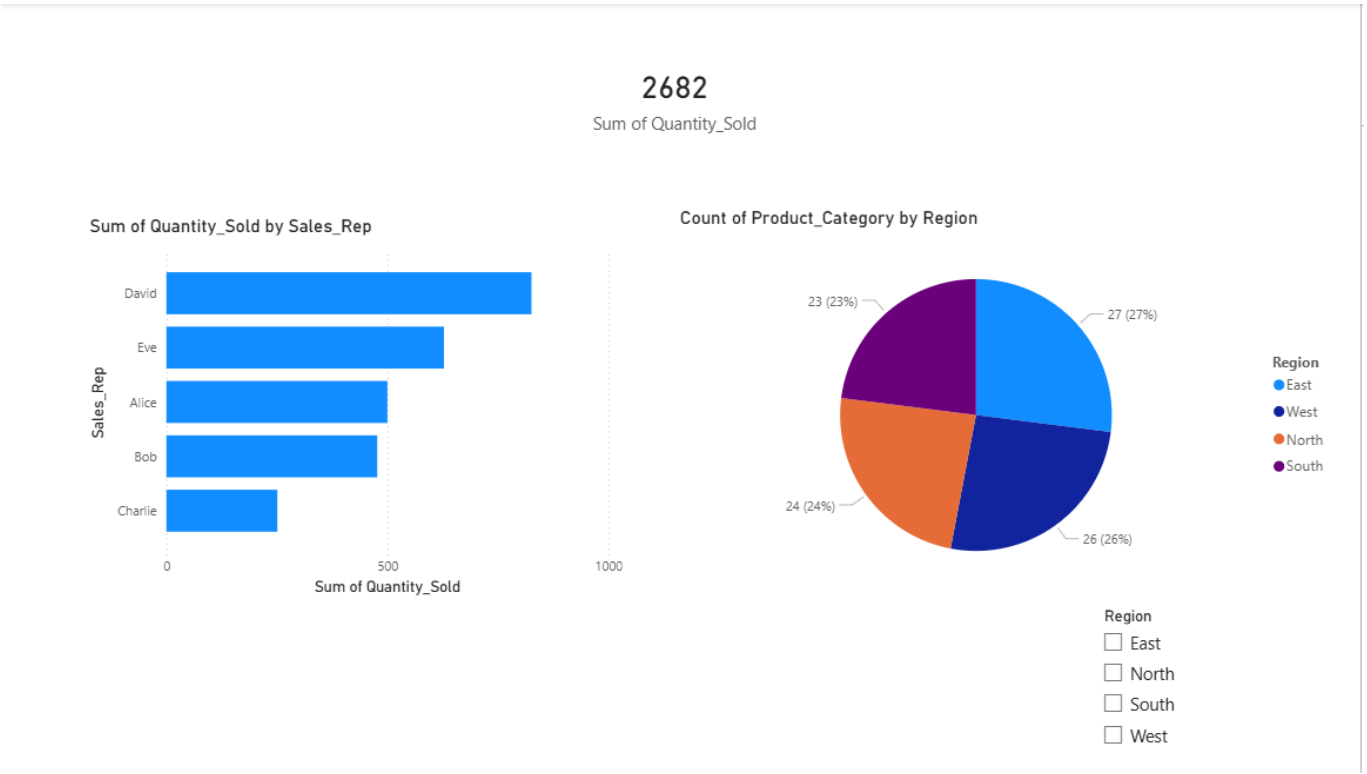
Assignment 5

1. Sales Data Analysis



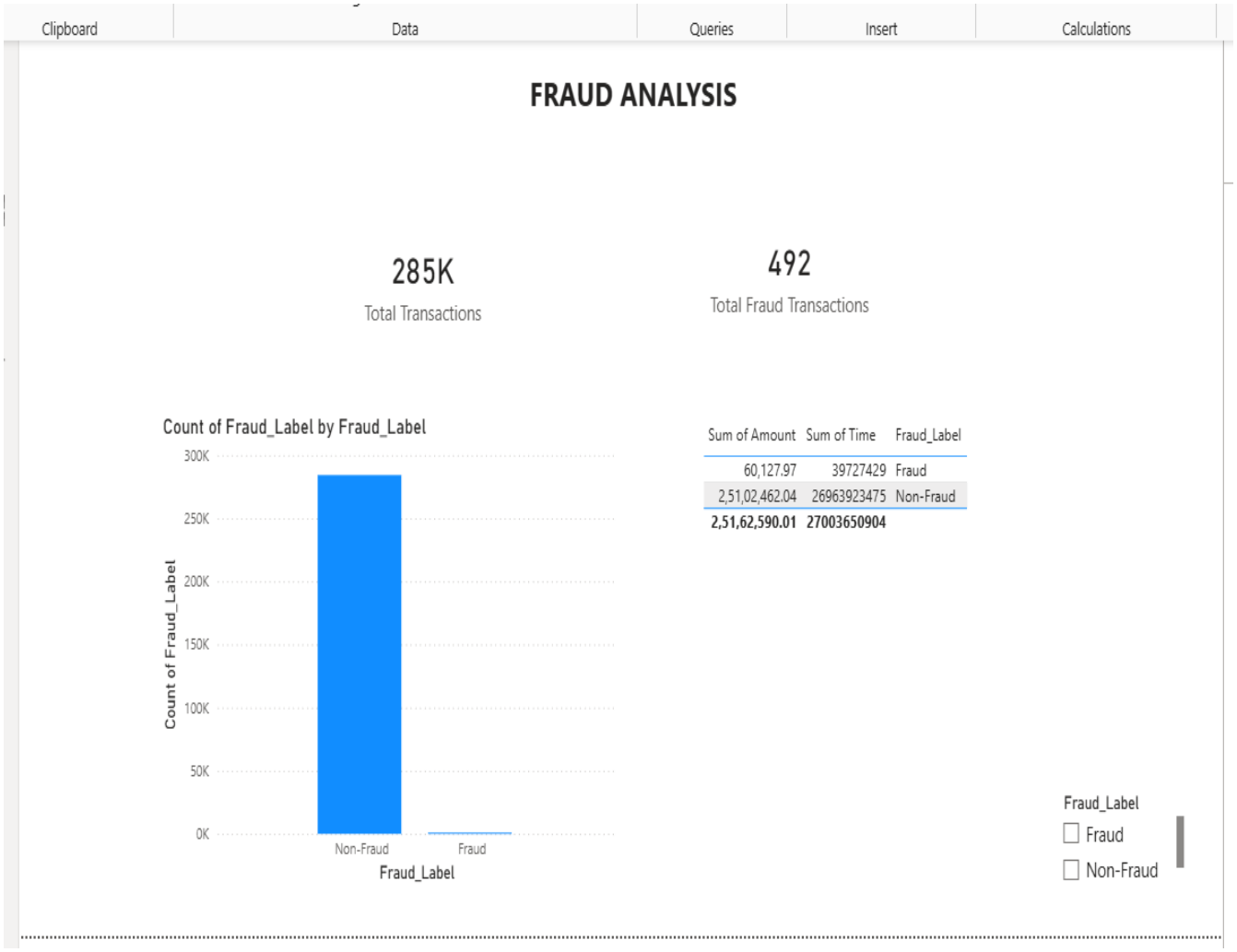
- The *Eldon File Cart* and *Rogers File Cart* products generate the highest sales, indicating strong demand.

- The *Central region* contributes the largest share of total sales, followed by *South* and *North*.

- Some products like *Index Tabs* and *Sharpeners* show very low sales, suggesting weaker demand.

- Regional and product-based insights help businesses *optimize inventory and focus on high-performing products*.

2. Fraud Detection Analysis



- The dataset contains around *285K transactions, but only **492 are fraudulent*, showing that fraud cases are very rare.

- The visualization highlights a *strong imbalance between fraud and non-fraud transactions*.

- Such imbalance makes fraud detection *challenging and requires specialized detection techniques*.

- Monitoring transaction patterns helps organizations *detect suspicious activity and reduce financial losses*.